

[RVHS/2025/Q4]

4 RV SC Election

Name your Jupyter Notebook as

Task_1_<class>_<index number>_<name>.ipynb

River Valley Hypothetical School Student Council had decided to use digital voting system for 2025 election. For 2025, the election will be for groups of 3 candidates where the student population will vote for their choice group. From the elected group, the President and 2 Vice Presidents will then be appointed.

To help manage the election process, RdeV is engaged to help with the programming of the database system. RdeV decided to use a relational database to store records of candidates' data, groups' data, and vote data. A separate table will store the candidate-group information as SC had indicated that it is possible for a candidate to be in different groups. The database will have 4 tables to store data on (1) **candidates**; (2) **groups**; (3) **candidate_group** and (4) **votes**. All fields cannot be left empty.

candidates:

- `candidate_id` – candidate identification. Example, `sc2025_01`
- `name` – name of candidate.
- `cls` – class for which the candidate belongs to. Example, `24J13`
- `gender` – gender of the candidate. Can only be `male` or `female`.
Database system will validate the value before accepting the record.

groups:

- `group_id` – group identification. Example, `1`.
- `name` – name of the group.

candidate_group:

- `candidate_id` – identification of candidate. It is possible that for the candidate to be in multiple groups
- `groups` – identification of groups. Each group will have 3 candidates.

votes:

- `voter` – student voters serial number. Example, `2025001`
- `vote` – group identification for which student voters will be casting their vote.

For each of the task 4.1 to 4.5, complete the codes in Jupyter Notebook and add a comment statement at the beginning of the code using the hash symbol `#`, to indicate the sub-task the program code belongs to, for example

In [1]:

```
#Task 4.1  
Program code
```

Output:

[RVHS/2025/Q4 Cont']

Task 4.1

Write a Python program that uses SQL code to create database `sc_election_2025.db` with the 4 tables given. The database is in the same directory as this Jupyter Notebook. Define the primary and foreign keys for each table and set up the necessary constraints in your codes. [6]

Task 4.2

A sample dataset was generated for the different tables for testing the database and the programs using it. The csv files `candidates.csv`, `groups.csv`, `candidate_group.csv` and `votes.csv` store the comma-separated values for each of the tables in the database corresponding to the table names.

Write a Python program to read in the data from each file and then store each item of the data in the correct place in the database. [4]

Task 4.3

After the voting process, the elected group name and candidate names need to be announced from the `group_id`.

Write a Python program that will print out the group name and the candidate names from a given `group_id`. The print out will follow the format of the following sample print out with the group name on the first line and the candidate names, in no order for the subsequent lines:

```
Student Voice -  
Lee Loong Loong  
D Dave  
Nnarnia Saladin
```

Run the program for `group_id 1`. [4]

Task 4.4

After the voting process, tabulation of the vote count for the different groups is needed to determine the winner of the election.

Write a Python program that will print out the vote count for the different groups in descending order of vote count. The print out will include the group's name and `group_id` in parenthesis and the vote count following the format of the sample print out:

```
Student Voice          (1) - 69  
Student Right          (2) - 62  
Total Student          (3) - 56  
RV Together            (5) - 55  
RV Steady              (4) - 54 [4]
```

Task 4.5

The election results will be implemented as a web application utilising python flask microframework.

You are tasked to write part of the web application to display the 2025 SC election results. Your web application will return a HTML page on the root route to display a table showing the vote count of the different groups by descending order of vote count followed by the details of the winning group.



2025 SC Election Results

Group	Votes
Student Voice (1)	109
Total Student (3)	105
Student Right (2)	103
RV Together (5)	97
RV Steady (4)	83

Winner of 2025 SC Election

Student Voice

1. Lee Loong Loong
2. D Dave
3. Nnarmia Saladin

The webpage must look similar to the one shown above with the following for the applied CSS rules:

- 1) The logo width will be 100 px with aspect ratio unchanged
- 2) Title, sub-title and winning group will use h1, h2 and h3 respectively
- 3) The table width and the section to display details of winning group will be 500 px.

Save your program code as

Task_4_5_<class>_<index>_<name>.py

With any additional files / subfolders as needed in a folder named

Task_4_5_<class>_<index>_<name>

Run the web application and save the returned pages in the root directory as

Task_4_5_<class>_<index>_<name>.html

[7]